Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech **Time: 03 Hours**
Code: 101404 **Subject: Introduction to Fluid Mechanics** **Full Marks: 70**

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) Liquids rise in capillaries due to
(i) Surface Tension
(ii) Diffusion
(iii) Osmosis
(iv) Viscosity

(b) A manometer is used to measure
(i) Atmospheric pressure
(ii) Pressure in pipes and channels
(iii) Pressure in Venturimeter
(iv) Difference of pressures between two points in a pipe

(c) In steady flow, streamlines, pathlines, and streaklines
(i) Are always different
(ii) Are always perpendicular
(iii) Coincide
(iv) Do not exist

(d) Which equation represents the continuity equation for incompressible, 3D flow?
(i) $\partial u/\partial x = \partial v/\partial y = \partial w/\partial z$
(ii) $\nabla . \vec{V} = 0$
(iii) $\nabla . \vec{V} = \rho$
(iv) $\nabla \times \vec{V} = 0$

(e) To apply Bernoulli's theorem, it is necessary for the fluids
(i) to be at atmospheric pressure
(ii) must be incompressible
(iii) to be of unit density
(iv) should have high viscosity

(f) Rotameter is a device used to measure
(i) Absolute pressure
(ii) Velocity of fluid
(iii) Flow
(iv) Rotation

(g) The boundary layer is formed as the when fluid comes in contact with solid surface boundary layers formed due to
(i) Surface tension
(ii) Forces of adhesion
(iii) Force of gravity acting on the fluid
(iv) Viscosity of the fluid

(h) The ratio of maximum velocity to average velocity in a fully developed laminar viscous flow in a pipe is
(i) 3.0
(ii) 2.0
(iii) 2.5
(iv) 1.5

(i) Which of the following is true for fully developed laminar flow in a circular pipe?
(i) Velocity varies along the pipe length
(ii) Velocity profile is linear
(iii) Pressure gradient is constant
(iv) Shear stress is zero throughout

(j) Principles of similitude form the basis of
(i) Performing acceptance tests
(ii) Comparing two identical equipments
(iii) Comparing similarity between design and actual equipment
(iv) Design and testing models of prototype based on results of models

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Q.2 (a) Define and mention units for the following fluid properties: Density, specific weight, specific volume and specific gravity of a fluid. [7]

(b) A 400 mm diameter shaft is rotating at 200 r.p.m in a bearing of length 120 mm. if the thickness of the oil film is 1.5 mm and the dynamic viscosity of the oil is 0.7 Pa.S. Determine the torque required to overcoming friction in bearing and power utilized in overcoming viscous resistance. Assume a linear velocity profile. [7]

Q.3 (a) Define metacentre and metacentric height. Derive an expression for metacentric height for experimental method. [7]

(b) A simple manometer (U-tube) containing mercury is connected to a pipe in which an oil of specific gravity 0.8 is flowing. The pressure in the pipe is vacuum. The other end of the manometer is open to the atmosphere. Find the vacuum pressure in pipe, if the difference of mercury level in the two limbs is 20 cm and height of oil in the left limb from the centre of the pipe is 15 cm below. [7]

Q.4 (a) Derive an expression for the depth of centre of pressure from free surface of liquid of an inclined plane surface submerged in the liquid. [7]

(b) Explain the three conditions of equilibrium developed when a floating body is given a slight angular displacement. [7]

Q.5 (a) Define a venturimeter and derive the equation of venturimeter to measure the discharge through a pipe. [7]

(b) Calculate the stream and velocity potential function for a 2-D flow field given by $u = 2x$ and $v = -2y$. [7]

Q.6 (a) Write the expression for acceleration of a fluid in x, y and z directions. Differentiate between local and convective acceleration. [7]

(b) Find out the rate of flow of an oil (S = 0.92) in a pipe with the help of an orifice meter whose diameter is 6 cm. The pipe diameter is 12 cm and pressure difference across the orifice meter is measured as 25 cm of mercury. Take the value of discharge coefficient as 0.60. [7]

Q.7 (a) Using Buckingham's π theorem prove that frictional torque T of a disc of diameter D rotating at a speed of N in a fluid of viscosity μ and density ρ in a turbulent flow is given by [7]

$$T = D^5 N^2 \rho \varphi \left[\frac{\mu}{DN\rho}\right]$$

(b) For the laminar flow through the circular pipe prove that shear stress variation across the pipe section is linear. [7]

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Q.8 (a) Define major and minor energy losses in pipes. Using Bernoulli's equation, obtain an expression for loss of head due to sudden expansion of pipe. [7]

(b) An aeroplane flying at a height of 15km, where the temperature is -50°C The speed of the plane is corresponding to Mach numbers is 2.0. Assuming k =1.4 and R = 287 J/kg K, find the speed of the plane. [7]

Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Vena-contracta and Cavitation
(b) Circulation and Vorticity
(c) Water hammer in pipes
(d) Froude Number, Mach Number, Weber Number and EulerNumber

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